In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("churn.csv")
print("Shape:", df.shape)
df.head()
```

```
Shape: (300, 8)
```

Out[2]:

	customer_id	gender	age	tenure	monthly_charges	total_charges	contract
0	1	Male	51	21	117.92	6825.09	Two year
1	2	FEMALE	56	28	28.16	6095.94	One year
2	3	Female	25	32	55.84	3074.01	Two year
3	4	Male	50	10	108.77	4773.68	Month-to-month
4	5	male	67	21	82.40	2290.87	Two year

In [3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   customer_id      300 non-null    int64  
 1   gender           300 non-null    str    
 2   age              300 non-null    int64  
 3   tenure           300 non-null    int64  
 4   monthly_charges  300 non-null    float64
 5   total_charges    300 non-null    float64
 6   contract         300 non-null    str    
 7   churn            300 non-null    str    
dtypes: float64(2), int64(3), str(3)
memory usage: 18.9 KB
```

In [4]:
```
print("Missing values in each column:")
print(df.isnull().sum())
print()
print("Total missing values:", df.isnull().sum().sum())
```

```
Missing values in each column:
customer_id        0
gender             0
age                0
tenure             0
monthly_charges    0
total_charges      0
contract           0
churn              0
dtype: int64

Total missing values: 0
```

In [5]:
```

```

```
print("Duplicate rows:", df.duplicated().sum())
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

```
Duplicate rows: 0
Shape after removing duplicates: (300, 8)
```

In [6]:
```
print("Unique gender values:", df["gender"].unique())
print("Unique contract values:", df["contract"].unique())
print("Unique churn values:", df["churn"].unique())
```

```
Unique gender values: <StringArray>
['Male', 'FEMALE', 'Female', 'male ']
Length: 4, dtype: str
Unique contract values: <StringArray>
['Two year', 'One year', 'Month-to-month']
Length: 3, dtype: str
Unique churn values: <StringArray>
['No', 'Yes']
Length: 2, dtype: str
```

In [7]:
```
df["gender"] = df["gender"].str.strip().str.capitalize()
print("Gender values after standardization:")
print(df["gender"].value_counts())
```

```
Gender values after standardization:
gender
Female    155
Male      145
Name: count, dtype: int64
```

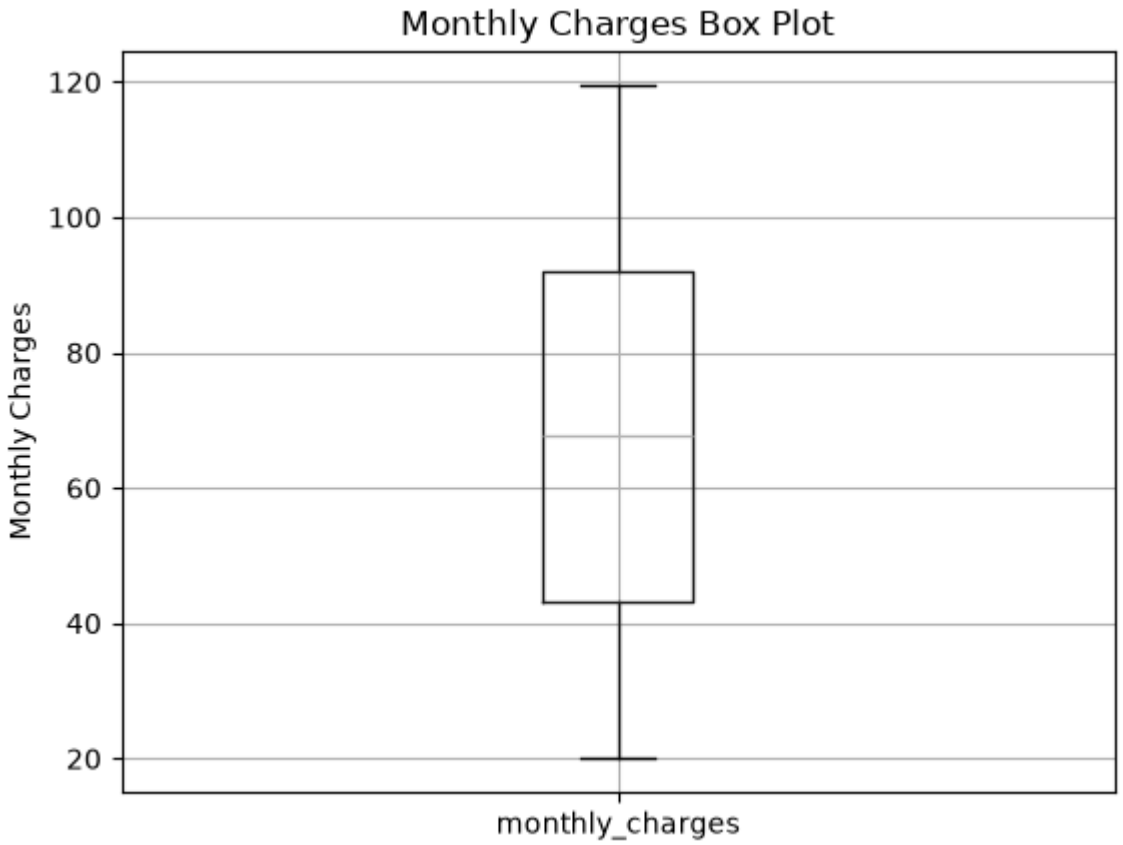
In [8]:
```
for col in ["age", "tenure", "monthly_charges", "total_charges"]:
    df[col] = df[col].fillna(df[col].median())
print("Missing values now:", df.isnull().sum().sum())
```

```
Missing values now: 0
```

In [9]:
```
q1 = df["monthly_charges"].quantile(0.25)
q3 = df["monthly_charges"].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = df[(df["monthly_charges"] < lower) | (df["monthly_charges"] >
print("Lower limit:", round(lower, 2), "| Upper limit:", round(upper, 2))
print("Outlier rows in monthly_charges:", len(outliers))
```

```
Lower limit: -30.54 | Upper limit: 165.68
Outlier rows in monthly_charges: 0
```

In [10]:
```
df.boxplot(column="monthly_charges")
plt.title("Monthly Charges Box Plot")
plt.ylabel("Monthly Charges")
plt.show()
```



In [11]:
```
df["avg_spend_per_month"] = df["total_charges"] / df["tenure"]
df["tenure_group"] = pd.cut(df["tenure"], bins=[0, 12, 36, 72], labels=["
df["churn_label"] = df["churn"].map({"No": 0, "Yes": 1})
df.head()
```

Out[11]:

	customer_id	gender	age	tenure	monthly_charges	total_charges	contract	chu
0	1	Male	51	21	117.92	6825.09	Two year	
1	2	Female	56	28	28.16	6095.94	One year	Y
2	3	Female	25	32	55.84	3074.01	Two year	
3	4	Male	50	10	108.77	4773.68	Month-to-month	Y
4	5	Male	67	21	82.40	2290.87	Two year	

In [12]:
```
print("Shape:", df.shape)
print("Missing values:", df.isnull().sum().sum())
print("Duplicate rows:", df.duplicated().sum())
print("Gender values:", df["gender"].unique())
print(df.dtypes)
```

```
Shape: (300, 11)
Missing values: 0
Duplicate rows: 0
Gender values: <StringArray>
['Male', 'Female']
Length: 2, dtype: str
customer_id               int64
gender                      str
age                       int64
tenure                    int64
monthly_charges         float64
total_charges           float64
contract                    str
churn                       str
avg_spend_per_month     float64
tenure_group           category
churn_label               int64
dtype: object
```

In [13]:
```python
df.to_csv("churn_cleaned.csv", index=False)
print("Cleaned data saved to churn_cleaned.csv")
saved = pd.read_csv("churn_cleaned.csv")
print("Saved file shape:", saved.shape)
saved.head()
```

```
Cleaned data saved to churn_cleaned.csv
Saved file shape: (300, 11)
```

Out[13]:

	customer_id	gender	age	tenure	monthly_charges	total_charges	contract	chu
0	1	Male	51	21	117.92	6825.09	Two year	
1	2	Female	56	28	28.16	6095.94	One year	Y
2	3	Female	25	32	55.84	3074.01	Two year	
3	4	Male	50	10	108.77	4773.68	Month-to-month	Y
4	5	Male	67	21	82.40	2290.87	Two year	